Speed and things

6a

The boxes below contain notes related to motion. Use information from these notes to help in answering the following questions.

Deceleration is negative acceleration. It occurs when an unbalanced force acts on an object to oppose its movement.

Acceleration is the change in velocity and can be calculated using the equation

$$a = (v_2 - v_1)/t$$

where v_2 = final velocity, v_1 = initial velocity and t = time taken.

Common units for speed are km/h or m/s.

Velocity is the measure of distance travelled in a unit time. The concept involves direction as well.

Speed is the distance travelled in a unit time.
speed = distance/time

Distance-time graphs can be used to find speeds and speed changes as well as distances travelled.

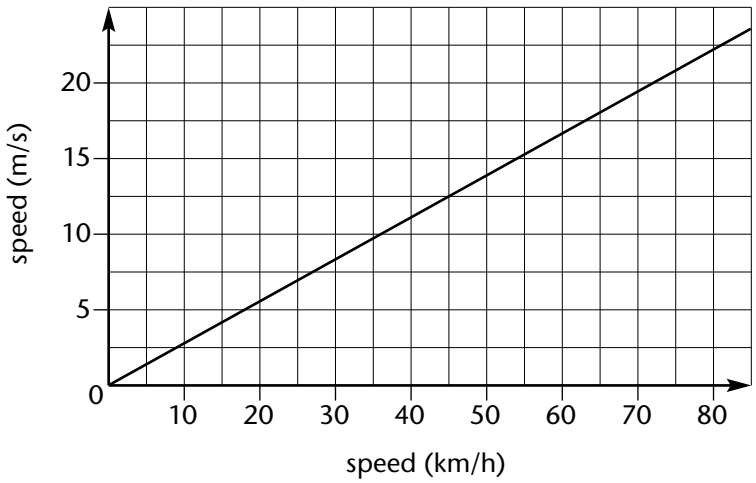
A common acceleration unit is metres per second per second (m/s/s or m/s^2).

In addition to showing speed and time, speed–time graphs can also indicate constant speed, acceleration and deceleration.

Friction is a force. A friction force always opposes the motion of bodies it acts upon.

All objects have inertia. The Law of Inertia, Newton's first law, states that objects remain stationary or continue to travel at a constant velocity – unless acted on by an unbalanced force.

Speed conversion graph



1 State Newton's first Law of Motion, the Law of Inertia.

_____

2 Why do we need to wear seat belts in a moving car?

_____

_____

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3 A passenger was standing at the front of the bus waiting to get out. The bus driver stopped suddenly, throwing the passenger forward. Explain what happened.

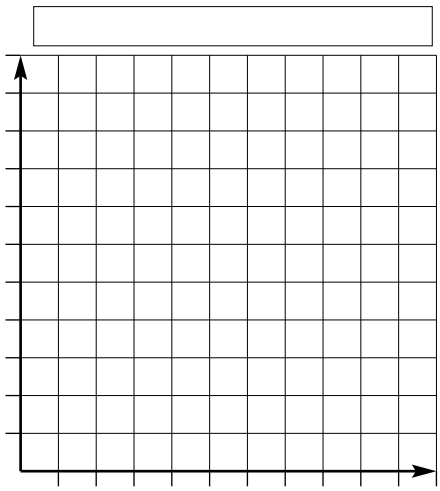
_____

_____

_____

4 Use the information from the table below to draw a speed–time graph. Add a title to your graph. (Hint: Remember, time goes on the horizontal axis.)

Time (s)	Speed (m/s)
1	10
2	10
3	10
4	7
5	5
6	3
7	4
8	6
9	8
10	8



5 Between what times was the car:

a moving at a constant speed? _____

b accelerating? _____

c decelerating? _____

6 A moving car accelerated uniformly for 12 seconds. Before it started to accelerate, its velocity was 6 m/s. When it finished accelerating, its velocity was 24 m/s. What was the car's acceleration?

_____

_____

7 A car was moving at 25 m/s and then quickly decelerated to 0 m/s in 5 seconds because a dog ran across the road. What was the car's:

a initial velocity? _____

b final velocity? _____

c acceleration? _____

d Is the answer to Question 7c positive or negative? Why?

_____

_____

8 **a** On the next page there are three velocity–time graphs. Indicate the graph (use the graph number) that represents a car:

– moving with constant velocity _____ – accelerating _____ – decelerating _____

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Speed and things

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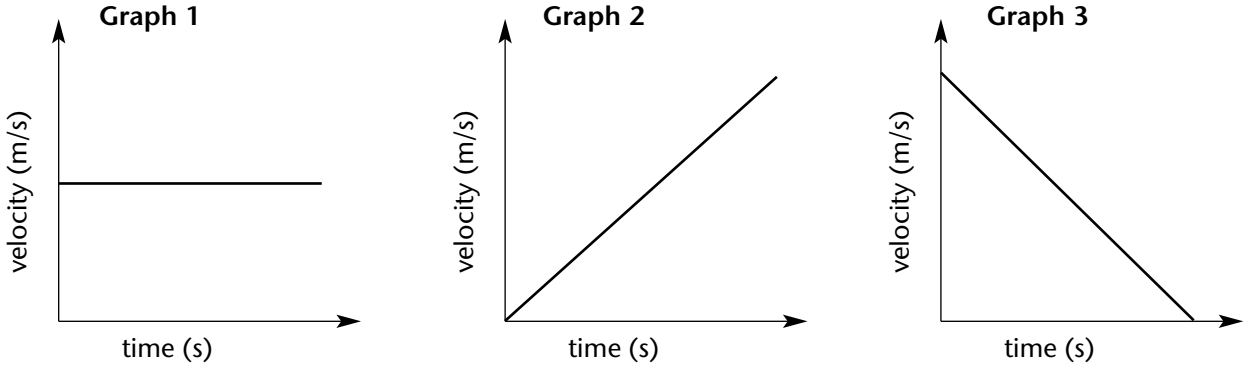

b Give a reason for each choice you made in part a. Identify the feature of each graph you used to make your selection.

9 A car was travelling at 18 m/s and then accelerated at 4 m/s^2 until its speed was 30 m/s. How long did this take? Show your working.

10 What is the difference between speed and velocity?

11 Given distance travelled and time taken to travel the distance, what is the equation for calculating speed?

12 State two uses for each of the following graphs:

a speed–time graphs

b distance–time graphs.

13 A car was driven to Brisbane, a distance of 1200 km, in 20 h. What was the average speed of the trip?

14 Express each of the following speeds in the units indicated in brackets:

a 60 km/h (m/s) _____

b 100 km/h (m/s) _____

c 200 m/s (km/h) _____

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Speed and things

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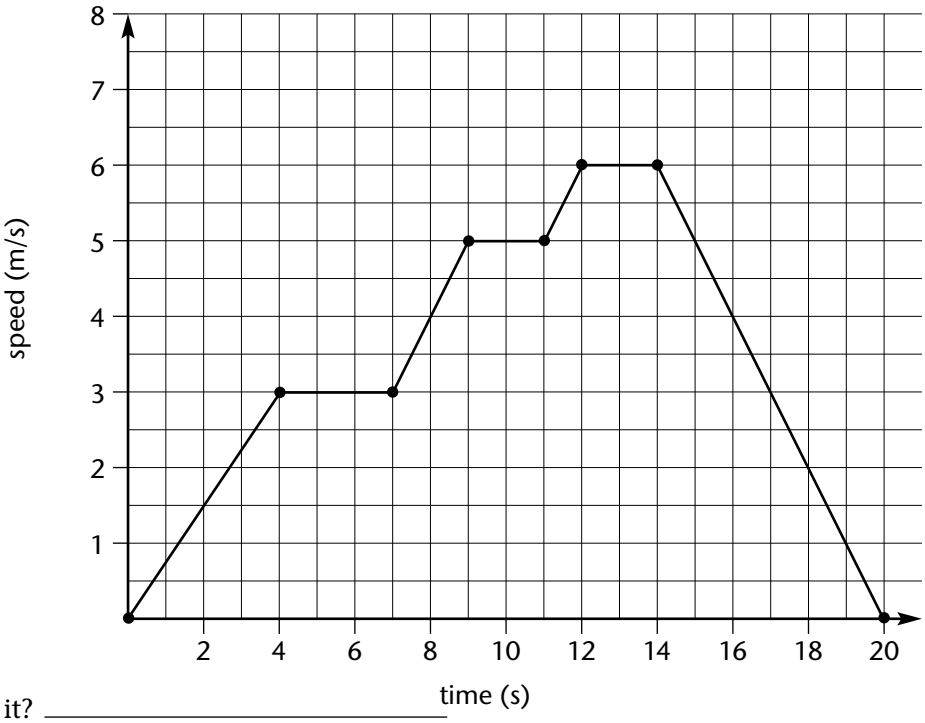
15 Complete the table below, which relates distances, times and average speeds. Show your working out.

Distance travelled	Time taken	Average speed	Calculation/working out
20 km	2 h		
1000 m	20 s		
50 km	5 h		
5000 m	5 min		
	2 h	60 km/h	
	30 s	200 m/s	
30 m		10 m/s	
600 km		60 km/h	

16 This graph represents the motion of a bike over a 20-second time period.



a What type of graph is it? _____

b During which time intervals was the bike travelling at constant speed?

_____

c What was the velocity of the bike at 12 seconds?

_____

d How long did it take the bike to reach a velocity of 5 m/s?

_____

e Fill in the table at right, showing the data from the marked points on the graph. Don't forget to label the columns.

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